

Anomaly Detection

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 18: AI Security Use Cases

1. The Detection Problem Anomaly Detection Was Built to Solve

Signature-based and rule-based detection tools are built to recognize known threats: they match observed activity against a catalogue of documented malicious patterns. This model worked reasonably well in an era when attackers used distinctive, custom malware and predictable techniques. Sophisticated modern adversaries exploit legitimate credentials, use built-in operating system tools (living off the land), and deliberately design their techniques to avoid matching known signatures. The 2024 Mandiant M-Trends report found a median global dwell time of 10 days between initial compromise and detection. In those 10 days, an attacker with initial access can establish persistence, move laterally, locate sensitive data, and begin exfiltration — all using activity that appears normal to rule-based tools. Anomaly detection takes a fundamentally different approach: rather than asking 'does this match something bad?', it asks 'does this deviate from what is normal for this specific environment and this specific entity?'

2. Building the Behavioral Baseline

Anomaly detection systems establish a behavioral baseline during an observation period, typically two to six weeks, before they begin producing reliable detections. During this phase the AI catalogs normal operating patterns: which users authenticate at which hours from which locations, which systems communicate with which peers on which ports, what volumes of data traverse specific network segments, which processes run on particular endpoint types. This baseline is environment-specific — it is not a global average but a model of what is normal for your organization. Rushing the training period is one of the most common and costly implementation mistakes: an AI system that has not adequately learned the baseline will produce excessive false positives, and analysts who receive too many false alerts quickly lose confidence in the system and begin ignoring its output.

Operational Rule: Never rush the baseline training period. An anomaly detection system that generates constant false positives will be tuned off or ignored — resulting in worse detection than no AI system at all.

3. NIST CSF 2.0 Alignment: The Detect Function

Anomaly detection sits at the core of the Detect function in NIST CSF 2.0, particularly under the DE.AE (Adverse Event Analysis) subcategory. Candidates should understand the specific CSF 2.0 subcategories that anomaly detection satisfies and the new addition in v2.0 that extends monitoring inward to AI systems themselves.

CSF 2.0 Subcategory	Requirement	How Anomaly Detection Satisfies It
DE.AE-01	Baseline of network operations and expected data flows established	Training period creates organization-specific behavioral baseline for all monitored entities

DE.AE-02	Potentially adverse events analyzed to understand associated activities	Anomaly scoring + session contextualization provides correlated analysis, not isolated event logging
DE.AE-06	Information on adverse events provided to authorized staff and tools	Anomaly alerts feed SIEM/SOAR integration for analyst workflow and automated response
DE.AE-08 (NEW in v2.0)	Detection of incidents affecting AI systems	ML observability tools (Arize, Fiddler, WhyLabs) apply anomaly detection to model behavior itself

4. User and Entity Behavior Analytics (UEBA)

UEBA platforms apply anomaly detection specifically to the behavioral patterns of users and non-human entities — service accounts, applications, devices. The critical innovation in modern UEBA is peer-group baselining: comparing individual behavior not only against that entity's own historical patterns but against the patterns of similar entities (users in the same role, department, or access tier). This catches anomalies that appear individually unremarkable but deviate from role norms. Microsoft Sentinel's UEBA module, Splunk UEBA, and Exabeam Fusion all implement peer-group clustering. Exabeam assigns risk scores to each session based on deviation magnitude from both personal and peer-group baselines, surfacing the highest-risk sessions for analyst prioritization.

Real-World Incident — Financial Services Insider Threat (2022): An employee in a treasury department systematically downloaded customer financial records to a personal cloud storage account over three months. Each individual download fell within normal per-session parameters — deliberately designed to evade threshold-based DLP alerts. AI-driven UEBA identified the cumulative pattern: total data volume over 90 days was 40 standard deviations above the employee's 18-month baseline for that data category. Security operations received an alert with a complete session timeline, enabling a forensic investigation that confirmed approximately 200,000 customer records had been copied. The employee had never exceeded any single-session access control threshold. Only behavioral anomaly detection identified the misuse across the full 90-day window.

5. Network Traffic Anomaly Detection

Network-layer anomaly detection platforms apply unsupervised machine learning to packet-level and flow-level traffic, building per-device communication models. When a workstation that has never initiated an SMB connection to a domain controller suddenly begins lateral movement behavior consistent with credential-dumping tools, the system flags it — not based on a signature match, but because the behavior deviates from the device's established communication model. This is the key differentiator from rule-based IDS: pattern-matching IDS can only fire when observed traffic matches a known malicious signature; anomaly detection fires when traffic deviates from established normal, catching novel techniques that have not yet been added to signature sets.

Platform	Layer / Focus	Key Capability	Notable Use Case
Darktrace	Network (packet-	Unsupervised ML; per-device	Identifies novel

	level)	communication model	lateral movement techniques without signatures
Vectra AI NDR	Network + Cloud + AD	Specific detection models for cloud and Active Directory attack patterns	Catches identity-based attacks in hybrid environments
AWS GuardDuty	Cloud API behavior	ML model of normal IAM and API call patterns	Detects IAM role misuse and unusual S3 operations
Microsoft Sentinel UEBA	User + entity	Peer-group risk scoring via Fusion ML	Correlates low-fidelity anomaly signals into high-confidence incidents
CrowdStrike Identity	Identity + temporal	Temporal anomaly scoring on authentication events	Flags credential compromise from off-hours, off-location logins

6. Temporal Context: Time as a Detection Feature

Time is one of the most powerful features in behavioral anomaly models because legitimate users exhibit highly predictable temporal patterns. A senior financial analyst who works Monday through Friday, 8 AM to 6 PM, and never logs in on weekends creates a strong temporal baseline against which weekend or late-night authentication events score as high anomalies — even when the login succeeds with valid credentials. CrowdStrike Identity Threat Protection and Microsoft Entra ID Protection both incorporate temporal anomaly scoring into identity risk calculations. This is particularly effective at detecting account compromise where attackers operate from a different time zone than the account owner, a scenario that evades most signature-based detections but registers prominently in temporal anomaly models.

7. Adversarial Evasion: How Attackers Defeat Anomaly Detection

Sophisticated adversaries specifically study anomaly detection to develop evasion strategies. BloodHound, the Active Directory reconnaissance tool widely used in red team engagements, can be configured to spread its LDAP queries across hours rather than minutes, keeping individual query rates within normal administrative traffic patterns. Nation-state actors who have spent time learning a target environment deliberately operate during business hours using the victim's own VPN to blend with legitimate user traffic. The defensive implication is important: anomaly detection tuned to fire only on individually unusual behaviors can be defeated by an attacker who understands the baseline. Correlating multiple weak signals simultaneously — unusual time of day, unusual data volume, unusual destination, unusual application, unusual device — raises detection probability significantly, even against deliberate evasion designed to defeat single-signal thresholds.

8. Cloud and SaaS Anomaly Detection

Cloud environments present unique anomaly detection challenges because the definition of 'normal' changes rapidly with autoscaling, ephemeral containers, and API-driven provisioning. AWS GuardDuty addresses this by modeling normal IAM API call patterns and flagging deviations — an IAM role that has historically performed only read operations suddenly issuing S3 PutBucketPolicy calls generates an immediate high-confidence alert. Google Cloud Security Command Center and Azure Defender for Cloud provide equivalent capabilities in their respective environments. SaaS-specific anomaly detection tools — Obsidian Security and Varonis for Microsoft 365, Google Workspace, and Salesforce — flag behavioral anomalies in productivity application activity, catching data exfiltration via legitimate SaaS channels that perimeter controls cannot observe.

9. System and Application Behavioral Anomalies

Anomaly detection extends beyond human users to non-human entities: servers, containers, application processes. A web server that has never spawned a child process and suddenly executes `cmd.exe` — a classic indicator of web shell compromise — is flagged immediately, even if the spawned process is not independently malicious. Application-level anomaly detection monitors metrics like authentication failure rate, response time, throughput, and data volume: a sudden spike in authentication failures on a login endpoint may indicate a credential stuffing attack in progress before any individual account is locked out. Microsoft Defender for Endpoint and CrowdStrike Falcon both monitor process trees at the kernel level, flagging parent-child process relationships that deviate from expected application behavior profiles.

10. Anomaly Detection for AI Systems Themselves

As AI models are deployed in production environments, they become security-relevant assets requiring the same behavioral monitoring applied to other infrastructure. Model behavior that drifts from its training distribution — producing unusual output distributions, receiving anomalous input patterns, or exhibiting unexpectedly high confidence on adversarially crafted inputs — may indicate active manipulation, adversarial attacks, or model degradation. This concept, called ML observability or model monitoring, is addressed in NIST AI RMF 1.0 under the GOVERN and MEASURE functions. Platforms including Arize AI, Fiddler, and WhyLabs provide ML observability capabilities that apply anomaly detection to model behavior over time. NIST CSF 2.0's new DE.AE-08 subcategory specifically addresses the detection requirement for incidents affecting AI systems — expect exam questions on this extension.

Memory Anchor — 'Monitor the Monitors': Just as you apply anomaly detection to user behavior and network traffic, you must apply it to the AI models doing the detecting. AI systems can be manipulated, drift, or degrade — and that requires the same behavioral baselines used for everything else.

11. SIEM, SOAR, and XDR Integration

Anomaly detection is most effective when it feeds an integrated response platform. SIEM platforms (Microsoft Sentinel, Splunk, IBM QRadar) aggregate anomaly alerts from UEBA, NDR, and EDR sources into correlated incidents, reducing alert volume through correlation. SOAR platforms (Palo Alto XSOAR, Splunk SOAR) automate initial response steps — isolating a compromised endpoint, disabling a

compromised account, blocking a suspicious IP — triggered by high-confidence anomaly events without requiring human intervention. XDR platforms (CrowdStrike Falcon XDR, Microsoft Defender XDR, SentinelOne Singularity) consolidate detection and response across all layers — endpoint, identity, network, cloud — into a single platform. An anomaly detection system whose output does not reach the SOC's primary working platform will not drive timely response, regardless of its technical sophistication.

12. False Positive Management

False positive rate is the primary operational challenge in anomaly detection programs. High false positive rates cause alert fatigue: analysts who receive too many incorrect alerts begin to ignore or suppress detection systems, ultimately producing worse security outcomes than a well-tuned simpler system. Common false positive sources include new employee onboarding (no baseline yet), legitimate user travel (unusual location for authorized activity), IT maintenance windows (authorized elevated activity), and business events like acquisitions that change normal communication patterns. Mitigation strategies include exception whitelisting for scheduled maintenance, new-user grace periods, and business event suppression windows. Peer-group baselining (Splunk UEBA, Exabeam) reduces individual-baseline false positives by grounding expectations in role-based norms rather than individual history alone.

Key Terms Glossary

Term	Definition
Anomaly Detection	A security approach that identifies statistical deviations from established behavioral baselines rather than matching against known malicious signatures.
Behavioral Baseline	A learned model of normal operating patterns for an entity (user, device, application, network segment) against which deviations are measured.
UEBA (User and Entity Behavior Analytics)	Anomaly detection applied to user activity and non-human entities (service accounts, applications), including peer-group baseline comparison.
Dwell Time	The period between initial compromise and detection; the 2024 Mandiant M-Trends global median was 10 days.
False Positive	An alert generated by anomaly detection for activity that is anomalous but legitimate; excessive false positives cause alert fatigue.
Alert Fatigue	Analyst desensitization caused by high volumes of low-fidelity alerts; leads to ignoring or suppressing detection systems.
Peer-Group Baselining	Comparing an entity's behavior against other entities with similar roles or characteristics, reducing false positives from individual-history-only baselines.
DE.AE-08 (NIST CSF 2.0)	New subcategory in CSF 2.0 requiring detection of incidents affecting AI systems themselves — extending anomaly detection to ML model behavior.
ML Observability	Application of behavioral anomaly detection to AI model outputs,

	input distributions, and confidence scores to detect manipulation, drift, or degradation.
Darktrace	Network anomaly detection platform using unsupervised ML to build per-device communication models at the packet level.
Exabeam Fusion	UEBA and SIEM platform that uses peer-group risk scoring to surface high-risk sessions for analyst review.
Temporal Anomaly	An anomaly detected based on the time at which activity occurs (e.g., authentication at 2 AM for a user with a strict 8 AM–6 PM baseline).
XDR (Extended Detection and Response)	A platform that consolidates detection and response across endpoint, identity, network, and cloud layers for cross-domain pattern correlation.

Quick Revision Checklist

- Anomaly detection: asks 'does this deviate from normal?' — complements signature detection which asks 'does this match known bad?'
- 2024 Mandiant M-Trends: global median dwell time = 10 days between compromise and detection.
- Baseline training period: 2-6 weeks; rushing it causes false positive-driven alert fatigue — do not skip.
- NIST CSF 2.0 DE.AE subcategories: DE.AE-01 (baseline established), DE.AE-02 (adverse event analysis), DE.AE-06 (alert distribution), DE.AE-08 NEW (AI system incident detection).
- UEBA: applies anomaly detection to users AND entities (service accounts, applications); includes peer-group baselining.
- Insider threat detection: cumulative behavior across extended periods (e.g., 90 days) catches slow-and-deliberate exfiltration that threshold-based tools miss.
- Network anomaly: Darktrace (per-device ML), Vectra AI NDR (cloud + AD), AWS GuardDuty (cloud API behavior).
- Temporal anomaly: time-of-day and day-of-week patterns are strong detection features; catches compromised accounts used by off-timezone attackers.
- Evasion: attackers operate during business hours, use victim VPN, spread activity to blend with baseline — correlate multiple weak signals to defeat evasion.
- Cloud SaaS: AWS GuardDuty, Azure Defender, Varonis, Obsidian Security — extend anomaly detection to API and SaaS activity.
- Application anomaly: web shell detection (web server spawning cmd.exe), credential stuffing (auth failure rate spike).
- ML observability: Arize AI, Fiddler, WhyLabs — apply anomaly detection to AI model behavior (drift, adversarial input, confidence anomalies).
- Integration chain: anomaly detection → SIEM (correlation) → SOAR (automated response) → XDR (cross-domain unification).
- False positive management: peer-group baselines, maintenance window suppression, new-user grace periods.

10 Exam-Based MCQs — Anomaly Detection

These questions reflect the scenario-heavy style of the CompTIA SecAI+ exam. Read every explanation carefully — understanding why each wrong option fails is as important as knowing the correct answer.

Q1. Scenario: A regional bank deployed a UEBA platform 8 months ago. The implementation team pressed to get detections running quickly and reduced the baseline observation period from the recommended 4 weeks to 5 days. The SOC manager reports that over the past 6 months, analysts have created a suppression rule that silences all UEBA alerts below high severity. Six months after deployment, the SOC manager reports that analysts have stopped triaging alerts from the UEBA platform because it generates hundreds of low-relevance alerts daily. What is the MOST likely root cause?

- A) The UEBA platform does not support peer-group baselining
- B) The baseline training period was too short, causing excessive false positives
- C) Anomaly detection is not appropriate for UEBA use cases
- D) The SIEM integration is blocking alerts from reaching analysts

Answer: B **Explanation:** B is correct because rushing the baseline training period — typically 2-6 weeks — results in an anomaly model that has not learned normal operating patterns. The underdeveloped model flags legitimate activities as anomalous, producing high false positive volumes that cause alert fatigue. When analysts stop trusting a system, they stop using it. A is wrong because while peer-group baselining helps reduce false positives, the primary cause described is insufficient training time, not missing feature. C is wrong because UEBA is specifically designed for user and entity behavior analysis — it is an appropriate use case. D is wrong because the symptom is too many alerts reaching analysts, not too few.

Q2. Scenario: Maya is a SOC analyst at a financial services firm. The UEBA platform has flagged an authentication event for the CFO's account at 11:30 PM Saturday. The CFO's baseline shows logins strictly between 7:30 AM and 7:30 PM, Monday through Friday, for the past 14 months. No other anomalies are associated with this session. A security analyst notices that a senior finance employee's account authenticated successfully at 11:30 PM on a Saturday from a known corporate IP. No other indicators of compromise are present. The analyst should FIRST:

- A) Dismiss the alert because the login used valid credentials from a known IP
- B) Escalate immediately as a confirmed compromise requiring account lockout
- C) Investigate the alert in the context of the user's temporal baseline and recent business events
- D) Submit a ticket to the network team to block the source IP address

Answer: C **Explanation:** C is correct because a single anomalous authentication event requires contextual investigation before action, not dismissal or immediate escalation. The analyst should verify: Does this user have a documented work-from-home arrangement? Is there a business event (quarter close, crisis) that explains weekend work? Has the account recently traveled? Temporal anomaly scoring is powerful but must be contextualized. A is wrong because valid credentials from a known IP do not rule out compromise — attackers who steal credentials use them from the victim's own infrastructure. B is wrong because immediately locking out a senior employee's account based on a single low-context anomaly risks business disruption and a false positive. D is wrong because blocking a known corporate IP would be inappropriate and likely cause collateral business impact.

Q3. Scenario: Marcus is a cloud security engineer at a SaaS company. An AWS GuardDuty alert fires at 2:14 AM for an IAM role used by the company's data pipeline application. The role's 6-month history shows exclusively `s3:GetObject` and `s3:ListBucket` calls against specific production data buckets. An AWS GuardDuty alert fires because an IAM role that has historically performed only read-only S3 operations suddenly called `S3 PutBucketPolicy`. The security team should treat this as:

- A) Expected behavior from a role with broad IAM permissions
- B) A high-priority anomaly indicating potential IAM role compromise or misconfiguration requiring immediate investigation
- C) A false positive because GuardDuty does not accurately model cloud IAM behavior
- D) Low-priority because `PutBucketPolicy` is not a destructive API call

Answer: B **Explanation:** B is correct because GuardDuty's ML-based behavioral model flags this because the role has never previously issued policy-modification calls. An IAM role suddenly acquiring write and policy-modification behaviors may indicate: the role's credentials were compromised and are being used for privilege escalation, a misconfigured application is making unintended API calls, or an insider threat is misusing an over-permissioned role. All scenarios warrant immediate investigation. A is wrong because broad IAM permissions do not explain a behavioral deviation — if the role has never used these permissions before, the sudden use is still anomalous. C is wrong because AWS GuardDuty is specifically designed and effective for cloud API behavior modeling. D is wrong because `PutBucketPolicy` can be used to make an S3 bucket public, remove access controls, or redirect access — this is a high-impact operation even without direct data destruction.

Q4. Which NIST CSF 2.0 subcategory specifically addresses the need to detect incidents affecting AI systems themselves?

- A) DE.AE-01
- B) DE.AE-02
- C) DE.AE-06
- D) DE.AE-08

Answer: D **Explanation:** D is correct. DE.AE-08 is a new subcategory introduced in CSF 2.0 (not present in v1.1) that explicitly requires detection capabilities for incidents affecting AI systems — extending the Detect function inward to cover AI/ML model behavior monitoring. A is wrong: DE.AE-01 addresses establishing a baseline of normal network operations and data flows. B is wrong: DE.AE-02 addresses analysis of potentially adverse events to understand associated activities. C is wrong: DE.AE-06 addresses providing information on adverse events to authorized staff and tools.

Q5. What distinguishes UEBA peer-group baselining from individual user baselining?

- A) Peer-group baselining uses signature matching instead of behavioral analysis
- B) Peer-group baselining compares behavior against similar users by role, reducing false positives from individual history anomalies
- C) Peer-group baselining requires a longer training period than individual baselining

D) Peer-group baselining is only applicable to service accounts, not human users

Answer: B **Explanation:** B is correct because peer-group baselining anchors anomaly scoring against the norms of similar users (same department, role, or access tier) rather than purely against individual history. This is valuable because new employees (no personal history), users returning from leave, and users whose roles have recently changed will all generate false positives from individual-only baselines even when their behavior is normal for their role. Peer-group context corrects this. A is wrong because peer-group baselining is a behavioral AI technique, not a signature matching approach. C is wrong because peer-group baselines converge with more data available from the group, not necessarily a longer calendar period. D is wrong because peer-group baselining applies to both human users and non-human entities.

Q6. A sophisticated attacker operating inside a network specifically spreads BloodHound LDAP queries across six hours to match normal administrative traffic rates. This technique is designed to evade:

- A) Signature-based IDS rules for BloodHound traffic
- B) Single-signal threshold-based anomaly detection
- C) Network firewall rules blocking LDAP queries
- D) Endpoint antivirus detection of the BloodHound binary

Answer: B **Explanation:** B is correct because the attacker is deliberately spreading activity to stay within single-session or single-hour thresholds that trigger anomaly alerts. By matching the rate of normal administrative LDAP traffic, the individual signals are each below anomaly thresholds. The defensive countermeasure is multi-signal correlation: combining unusual time-of-day context, cumulative volume over the session or day, unusual source-to-destination communication pair, and application identity to raise detection probability against deliberate baseline blending. A is wrong because spreading the queries affects rate-based anomaly detection, not signature matching — if BloodHound's specific query syntax is in an IDS signature, rate spreading does not help. C is wrong because LDAP queries to domain controllers are legitimate and would not be blocked by a well-configured firewall. D is wrong because this is a network and behavior evasion technique, unrelated to binary detection on endpoints.

Q7. An organization deploys Arize AI to monitor its machine learning fraud detection model. This is an example of:

- A) Applying SOAR automation to reduce alert response times
- B) ML observability — using anomaly detection to monitor AI model behavior for drift or adversarial manipulation
- C) Replacing human fraud analysts with automated detection
- D) Implementing MITRE ATT&CK-aligned detection for financial fraud patterns

Answer: B **Explanation:** B is correct because Arize AI is an ML observability platform that applies anomaly detection principles to model behavior — monitoring output distributions, input feature distributions, and confidence scores over time. Unusual patterns in these metrics may indicate model drift (degraded performance due to changed real-world data), adversarial manipulation (attackers crafting inputs to evade the model), or data pipeline failures. This is exactly the use case addressed by NIST CSF 2.0 DE.AE-08 and NIST AI RMF 1.0 GOVERN and MEASURE functions. A is wrong because Arize AI is a model monitoring tool, not a SOAR automation platform. C is wrong because ML observability augments analyst

capability — it does not replace human judgment in fraud investigation. D is wrong because ATT&CK alignment relates to threat actor technique mapping in security operations, not ML model behavioral monitoring.

Q8. What is the PRIMARY purpose of Microsoft Sentinel's Fusion ML capability in the context of anomaly detection?

- A) To train behavioral baselines for user and entity activity
- B) To correlate multiple low-fidelity anomaly signals from different data sources into high-confidence incident alerts
- C) To automatically remediate compromised user accounts without analyst intervention
- D) To generate MITRE ATT&CK-aligned threat reports from log data

Answer: B **Explanation:** B is correct because Fusion is Microsoft Sentinel's ML-based correlation engine that synthesizes low-confidence signals from disparate data sources — endpoint EDR, identity anomalies, network anomalies, cloud access logs — into correlated high-confidence incident alerts. Rather than presenting analysts with hundreds of individual low-severity anomaly events, Fusion surfaces a small number of incidents with full context about which data sources contributed, why the AI considers them related, and what response actions are recommended. This dramatically reduces analyst triage burden. A is wrong because baseline training is a function of the underlying UEBA module, not Fusion specifically. C is wrong because automated account remediation is a SOAR function, not Fusion's purpose. D is wrong because ATT&CK reporting is a separate capability from Fusion's correlation function.

Q9. Which anomaly detection approach is MOST effective for identifying a compromised executive account being used by an attacker operating from a different time zone?

- A) Network traffic anomaly detection using Darktrace
- B) Application error rate monitoring
- C) Temporal anomaly scoring on authentication events
- D) Signature-based IDS rules for known credential theft tools

Answer: C **Explanation:** C is correct because executives with predictable work schedules create strong temporal baselines — consistent login hours over months establish a reliable pattern. When an attacker in a different time zone uses the compromised account, authentication events will occur at times that deviate significantly from this baseline, generating high anomaly scores even when credentials are valid and MFA has been passed (if the attacker has also compromised MFA). CrowdStrike Identity Threat Protection and Microsoft Entra ID Protection both implement temporal anomaly scoring specifically for this use case. A is wrong because network traffic anomaly detection identifies deviations in communication patterns, not authentication timing — the attacker's network behavior may be indistinguishable from the executive's if using a VPN. B is wrong because application error rate monitoring detects service availability and reliability issues, not identity compromise. D is wrong because signature-based IDS cannot detect credential misuse that uses legitimate authentication pathways without triggering known tool signatures.

Q10. Which of the following BEST describes why anomaly detection and signature-based detection are complementary rather than competing capabilities?

- A) Anomaly detection is faster to execute than signature matching at scale
- B) Signature detection catches known threats efficiently; anomaly detection catches novel and unknown threats not yet in signature databases
- C) Anomaly detection replaces signature detection for all threats once the baseline is established
- D) Signature detection requires no training period while anomaly detection requires years of data

Answer: B **Explanation:** B is correct because the two approaches have complementary coverage profiles. Signature/rule-based detection is fast, precise, and produces no false positives for previously unseen threats — it either matches a known pattern or it does not. It misses novel attacks and variants that slightly modify known techniques to avoid matching. Anomaly detection catches statistical deviations regardless of whether the technique is documented, covering the novel and unknown at the cost of some false positive rate. Running both in parallel provides coverage that neither approach alone achieves. A is wrong because signature matching is typically faster computationally than ML-based anomaly scoring — speed is not the differentiator. C is wrong because anomaly detection explicitly cannot replace signature detection for known threats where precision and speed matter. D is wrong because anomaly detection baselines are typically established in 2-6 weeks, not years; signature detection does require no training but requires continuous database updates.